EduGraph AI

Offline AI-Powered Academic Performance, Assessment Verification, and Learning Intelligence Platform

Final Project Description

EduGraph AI improves academic performance by connecting curriculum standards, teacher-prepared assessments, student results, learning gap analysis, personalized study plans, and offline AI academic support.

1. Short Description

EduGraph AI is an offline-first AI education intelligence platform that helps schools improve student academic performance. Students first create a profile and select their future career interest. Teachers then upload their own prepared exams or tests, and the system verifies those assessments against the official curriculum and Curriculum Learning Outcomes (CLOs).

After students take the verified exams, EduGraph AI analyzes their performance, identifies learning gaps, detects weak topics and prerequisite weaknesses, and generates personalized academic study plans. The system also provides offline AI learning support through a school-based local server, helping students improve their grades and prepare for the academic requirements of their future career path.

At scale, EduGraph AI can support schools, regional education bureaus, and the Ministry of Education by providing evidence-based analytics on curriculum coverage, assessment quality, student performance, school quality, and national learning gaps.

2. Core Problem

Many students perform poorly not because they are incapable, but because schools often lack systems that clearly identify what students do not understand, what prerequisite knowledge is missing, whether exams align with curriculum standards, and what each student should study next.

Teachers prepare exams, but schools may not have strong tools to verify whether those exams fully align with curriculum standards and CLOs. Students receive marks, but often do not receive clear, personalized guidance on how to improve.

3. Main System Flow

```
Student selects career interest
  -> System understands priority subjects for that career
  -> Teacher prepares and uploads exam/test
  -> System verifies exam against curriculum and CLOs
  -> Students take the verified exam
  -> System analyzes results
  -> Knowledge graph identifies learning gaps
  -> System generates personalized academic study plan
  -> Student uses offline AI assistant to study
  -> Teacher and school monitor progress
  -> Regional and Ministry dashboards receive aggregated insights
```

The main goal is academic performance improvement. Career interest is used as a motivation and prioritization signal, not as the center of the product.

4. Key Users

User	Main value
Student	Selects career interest, takes assessments, sees learning gaps, follows study plans, uses offline AI tutoring, and improves grades.
Teacher	Uploads prepared exams, verifies CLO alignment, reviews class gaps, groups students by weakness, and recommends interventions.
School Admin	Monitors curriculum coverage, assessment quality, student performance, weak departments, and school improvement reports.
Regional Bureau	Compares schools, identifies regional subject weaknesses, monitors curriculum implementation, and recommends support.
Ministry of Education	Monitors national education quality, curriculum effectiveness, assessment quality, learning gaps, and policy needs.

5. Main Components

5.1 Student Profile and Career Interest

When a student first joins the system, they create a profile and select a career interest. The system then builds an academic priority map. For Electrical Engineering, Mathematics and Physics may become high-priority subjects. For Medicine, Biology and Chemistry may become high-priority subjects. However, the system still emphasizes overall academic performance across all required high school subjects.

5.2 Curriculum and CLO Knowledge Graph

The knowledge graph connects grades, subjects, units, topics, subtopics, CLOs, prerequisite concepts, difficulty levels, skills, and career relevance. This allows the system to understand what a question measures and why a student may have failed.

```
Grade 11 Physics
  -> Electricity
    -> Electric Circuits
      -> CLO: Analyze current, voltage, and resistance in a circuit
      -> Prerequisite: Algebra
      -> Prerequisite: Unit conversion
      -> Career relevance: Electrical Engineering
```

5.3 Teacher Exam Upload

Teachers prepare their own exams and upload them to EduGraph AI. The platform can support Word documents, PDFs, typed questions, spreadsheets, manual entry, and later scanned exams. Teachers provide grade, subject, unit, term, exam type, total marks, and intended CLOs if available.

5.4 AI-Assisted Exam Verification

EduGraph AI evaluates teacher-prepared exams against the official curriculum and CLOs. It checks question-to-CLO mapping, topic coverage, missing or over-tested CLOs, out-of-curriculum questions, difficulty balance, question clarity, and whether the assessment measures understanding rather than only memorization.

```
Example verification output:
Curriculum alignment: 86%
CLO coverage: Good
```

```
Difficulty balance: Moderate
Out-of-curriculum questions: 1
Under-covered topic: Magnetism
Recommendation: revise unclear questions and add missing CLO coverage.
```

5.5 Verified Assessment Repository

Verified assessments are stored with teacher, school, subject, grade, CLO mapping, topic mapping, difficulty level, verification score, approval status, student attempts, and results. Over time, this creates an institutional assessment memory for the school.

5.6 Student Examination System

Students can take exams digitally using devices connected to the school local Wi-Fi. For paper-based schools, teachers can print verified exams and later enter marks or upload answer data, allowing the system to still analyze gaps and generate plans.

5.7 Learning Gap Analysis

After the exam, the system analyzes performance beyond total marks. It identifies weak subjects, weak topics, failed CLOs, prerequisite gaps, repeated weakness patterns, and career-priority weaknesses.

```
Instead of only: Student scored 58%
EduGraph AI shows:
- Weak in algebra-based calculations
- Weak in unit conversion
- Weak in electric circuits
Likely root cause: weak prerequisite knowledge in algebra and formula application
```

5.8 Personalized Academic Study Plan

The system generates an academic study plan for each student. The plan focuses first on improving grades and curriculum mastery. Career interest helps prioritize some subjects but does not replace the need for strong overall performance.

```
Student goal: Electrical Engineering
Current academic average: 68%
Target academic average: 85%
Priority improvement areas: Mathematics and Physics
This week: review algebra, practice trigonometry, study Ohm's Law, complete a mastery
quiz
```

5.9 Offline AI Academic Assistant

Offline AI access is mainly for student academic support. Each school can deploy an EduGraph School Box that works through local Wi-Fi without internet. Students can ask why they failed a topic, request explanations, get practice questions, and follow their study plan.

The offline AI answers using curriculum content, CLOs, student exam results, learning gaps, career interest, personalized study plans, and approved school learning materials. It is a curriculum-aware academic assistant, not a generic chatbot.

5.10 Teacher Dashboard

Teachers see exam verification results, CLO coverage, topic coverage, student performance, class weak areas, students grouped by learning gap, recommended remedial actions, and study plan progress.

5.11 School Dashboard

School leaders see subject performance trends, teacher assessment quality, curriculum coverage, CLO mastery by grade, weak subjects, student improvement trends, intervention progress, and school quality reports.

5.12 Regional and Ministry Analytics

When multiple schools use EduGraph AI, selected data can be aggregated to regional and Ministry dashboards. Leaders can compare schools, identify weak subjects, detect curriculum implementation problems, monitor national learning outcomes, and support policy decisions.

6. Offline-First Architecture

```
Student / Teacher Devices
  -> Local Wi-Fi
    -> EduGraph School Box
      -> Web application
      -> Curriculum/CLO knowledge graph
      -> Student performance database
      -> Assessment verification engine
      -> Offline AI academic assistant
      -> Learning content repository
      -> Study plan engine
      -> Teacher dashboard
      -> School analytics dashboard
```

When internet is available, selected data can sync upward from school to regional bureau and Ministry dashboards. Student privacy should be protected, and higher-level dashboards should mainly use aggregated data unless authorized.

7. MVP Scope

The first version should be the EduGraph AI School Box MVP. It should focus on teacher assessment verification, student gap analysis, personalized academic study planning, and offline AI academic support.

- Student profile with career interest
- Curriculum/CLO setup for one subject
- Teacher exam upload
- AI-assisted exam verification
- Student exam-taking module
- Result storage
- Learning gap analysis
- Personalized academic study plan
- Offline AI academic assistant
- Teacher dashboard

Suggested first pilot: one school, one grade, one subject, and one unit - for example, Grade 11 Physics: Electricity.

8. What Not to Build First

- Full national Ministry dashboard
- All subjects and all grades
- Employer or industry marketplace
- Complex career matching
- Automated exam generation as the main feature
- Handwritten answer grading
- Full mobile applications
- Full national data integration

9. Final Value Proposition

EduGraph AI helps schools improve academic performance by connecting curriculum standards, teacher-prepared assessments, student results, learning gap analysis, personalized study plans, and offline AI tutoring. Students receive guidance based on both their current academic weaknesses and their future career interests, while teachers, schools, regions, and the Ministry gain evidence-based insights into education quality and learning outcomes.

10. Final Vision

EduGraph AI aims to become an offline-first education intelligence infrastructure that helps students improve grades, helps teachers improve assessment quality, helps schools improve learning outcomes, and helps education leaders make data-driven decisions.

```
Curriculum
  -> Assessments
    -> Student performance
      -> Learning gaps
        -> Personalized study plans
          -> Offline AI support
            -> School quality
              -> Regional improvement
                -> Ministry-level education intelligence
```

Source Basis

This description is based on the uploaded EduGraph AI innovation proposal and the uploaded AI-Powered National Education Quality and Performance Management System document. The first source describes the offline educational intelligence platform, knowledge graphs, student digital twin, learning gap prediction, and school analytics. The second source describes curriculum compliance, assessment verification, school quality assessment, student performance monitoring, and regional/Ministry-level analytics.